Mutual Fund Analytics Platform

Capstone Project Report

Bluestock Fintech Internship

Prepared By :

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Technologies Used :

- Python
- Pandas
- NumPy
- SQLite
- SQL
- Jupyter Notebook
- Power BI
- Git & GitHub

Project Repository :

github.com/rohitmurthy2803/mutual_fund_analytics

Executive Summary :

The Mutual Fund Analytics Platform is an end-to-end analytics solution developed to analyze mutual fund performance, investor behaviour, SIP trends, portfolio allocation and risk metrics. The project integrates data engineering, database management, exploratory data analysis, performance analytics and business intelligence into a unified workflow.

Ten mutual fund datasets were collected, cleaned and transformed using Python. The processed datasets were loaded into a SQLite database for structured querying and analysis. Exploratory Data Analysis was performed to identify trends in Assets Under Management (AUM), Systematic Investment Plan (SIP) inflows, investor participation, category-wise fund performance and portfolio allocation patterns.

Advanced financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Tracking Error, Value at Risk (VaR) and Conditional Value at Risk (CVaR) were implemented to evaluate risk-adjusted fund performance.

An interactive Power BI dashboard was developed to visualize industry trends, fund performance, investor analytics and market insights. The project demonstrates the practical application of

data analytics and business intelligence techniques in the mutual fund domain.

Problem Statement :

The mutual fund industry generates large volumes of financial and investor-related data. Fund houses, analysts and investors require efficient analytical tools to evaluate scheme performance, identify investment opportunities and monitor risk exposure.

Traditional analysis methods often focus only on absolute returns and fail to incorporate risk-adjusted performance indicators. Similarly, investor behaviour, SIP continuity and portfolio concentration are frequently overlooked despite their importance in long-term investment planning.

The objective of this project is to build a comprehensive Mutual Fund Analytics Platform capable of processing large datasets, performing advanced analytics and presenting actionable insights through interactive dashboards.

Project Objectives :

The primary objective of this project is to design and develop a comprehensive Mutual Fund Analytics Platform that enables efficient analysis of mutual fund performance, investor behaviour and market trends.

Specific objectives include:

- Build a structured data pipeline for mutual fund datasets.
- Clean and validate financial and investor data.
- Design and implement a relational database using SQLite.
- Perform exploratory data analysis to identify trends and patterns.
- Calculate performance and risk-adjusted metrics.
- Evaluate investor participation and SIP continuity.
- Develop an interactive Power BI dashboard.
- Generate actionable insights and recommendations for investors.

The project aims to bridge the gap between raw financial data and business decision-making by transforming data into meaningful insights.

Data Sources

The project utilizes ten datasets covering mutual fund performance, investor transactions, portfolio allocation and benchmark market indices.

Datasets Used :

1. Fund Master

Contains fund-level information such as AMFI codes, scheme names and categories.

2. NAV History

Historical Net Asset Value records used for return calculations and trend analysis.

3. AUM by Fund House

Assets Under Management information for various Asset Management Companies (AMCs).

4. Monthly SIP Inflows

Monthly SIP contribution data used to analyze investor participation.

5. Category Inflows

Category-wise investment inflows across different mutual fund segments.

6. Industry Folio Count

Investor folio statistics representing participation levels.

7. Scheme Performance

Returns, risk metrics and expense ratios for mutual fund schemes.

8. Investor Transactions

Detailed transaction records including SIPs, purchases and redemptions.

9. Portfolio Holdings

Portfolio allocation and holding weights used for concentration analysis.

10. Benchmark Indices

Market benchmark data used for performance comparison and risk calculations.

These datasets collectively provide a comprehensive view of mutual fund operations and investor activity.

Project Architecture

The Mutual Fund Analytics Platform follows a layered architecture consisting of data ingestion, processing, storage, analytics and visualization components.

Architecture Flow

Raw Data Sources

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Data Ingestion

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Data Cleaning and Validation

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SQLite Database

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SQL Analysis

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Exploratory Data Analysis

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Performance Analytics

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Advanced Risk Analytics

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Power BI Dashboard

Benefits of the Architecture

- Modular design
- Easy maintenance
- Scalable analytics workflow
- Reusable data pipeline
- Efficient reporting and visualization

The architecture ensures a clear separation between data engineering, analytics and visualization layers.

ETL Pipeline

The ETL (Extract, Transform, Load) process forms the backbone of the analytics platform.

Extraction

Raw datasets were collected from multiple sources and imported into Python using the Pandas library. Historical NAV data and benchmark information were used for financial calculations.

Transformation

Several data cleaning operations were performed:

- Missing value handling
- Duplicate record removal
- Date standardization
- Data type conversion

- Validation of financial metrics
- Transaction data verification

Dedicated cleaning scripts were developed:

- clean_nav_history.py
- clean_investor_transactions.py
- clean_scheme_performance.py

A master execution script (run_pipeline.py) was created to automate the ETL workflow.

Loading

The cleaned datasets were loaded into a SQLite database using structured schemas and SQL scripts.The ETL pipeline ensured consistency, reliability and analytical readiness of the data.

Database Design

SQLite was selected as the database management system due to its lightweight architecture and ease of integration with Python.

Database Components

Schema Design

A relational schema was created to store mutual fund information, performance metrics, investor transactions and benchmark data.

Tables Created

- fund_master
- nav_history
- aum_by_fund_house
- monthly_sip_inflows
- category_inflows
- industry_folio_count
- scheme_performance
- investor_transactions
- portfolio_holdings
- benchmark_indices

SQL Operations

Several SQL queries were developed to:

- Aggregate fund performance metrics
- Analyze SIP inflows
- Compare benchmark performance
- Evaluate investor participation
- Generate analytical summaries

The database layer provided a centralized and structured repository for all project datasets.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure, quality and characteristics of the mutual fund datasets. The analysis focused on identifying trends, patterns and relationships within the data before performing advanced analytics.

Key areas analyzed included:

- NAV trends across mutual fund schemes
- Assets Under Management (AUM) growth
- SIP inflow patterns
- Category-wise investment inflows
- Investor participation through folio counts
- Sector allocation across portfolios

More than 15 visualizations were created using Python and Matplotlib to support data-driven insights. The analysis helped identify growth patterns, investment preferences and fund performance trends across the industry.

EDA Findings

Several important insights were obtained during exploratory analysis:

- Industry AUM showed consistent growth during the analysis period.
- SIP inflows demonstrated increasing retail investor participation.
- Certain mutual fund categories attracted significantly higher inflows than others.
- Investor folio counts increased steadily, indicating industry expansion.
- NAV trends highlighted both market growth phases and correction periods.
- Portfolio allocation revealed varying levels of sector concentration among schemes.

The findings provided a strong foundation for subsequent performance and risk analytics.

Performance Analytics

Performance analytics was conducted to evaluate mutual fund returns and compare schemes using standardized financial metrics.

The following metrics were calculated:

- Daily Returns
- CAGR (1 Year, 3 Year and 5 Year)
- Sharpe Ratio
- Sortino Ratio

Daily returns were derived from historical NAV data and served as the basis for further calculations. CAGR was used to evaluate long-term growth performance, while Sharpe and Sortino Ratios measured risk-adjusted returns.

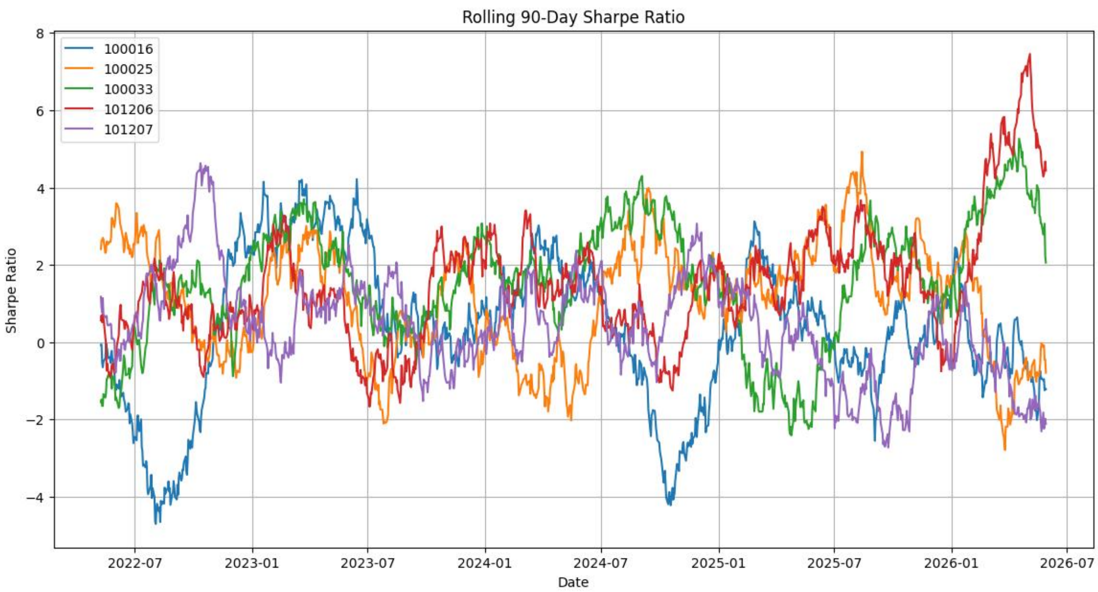
These metrics enabled objective comparison and ranking of mutual fund schemes.

Risk Metrics

Risk analysis plays an important role in mutual fund evaluation. Multiple risk indicators were calculated to understand downside exposure and volatility.

Metrics implemented include:

- Alpha
- Beta
- Maximum Drawdown
- Tracking Error
- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)



Alpha and Beta were calculated using benchmark index data. Maximum Drawdown measured peak-to-trough losses, while

Tracking Error quantified deviation from benchmark performance.

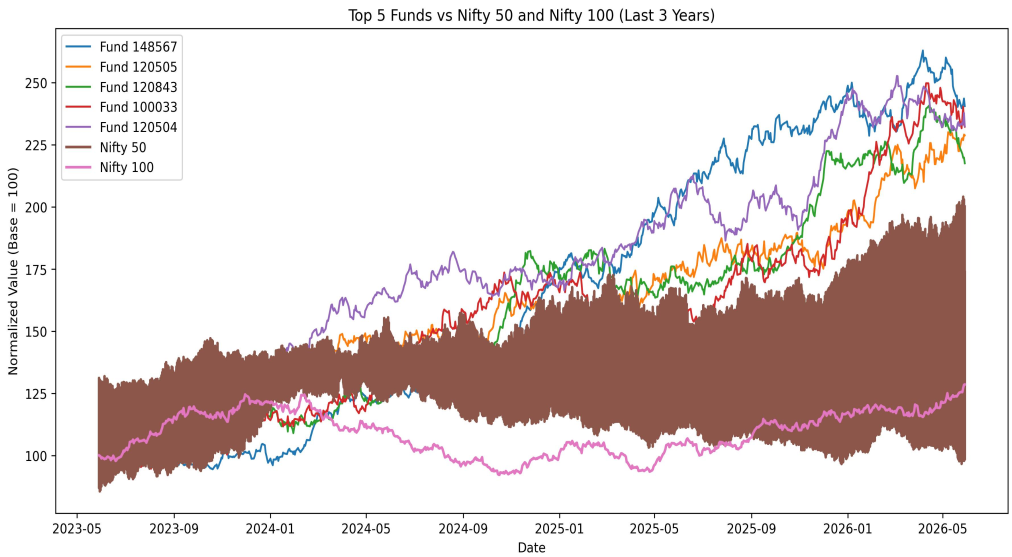
VaR and CVaR were used to estimate potential losses under adverse market conditions and assess downside risk exposure.

Fund Scorecard and Benchmark Analysis

A composite Fund Scorecard was developed to rank mutual fund schemes based on multiple performance and risk indicators.

The scorecard incorporated:

- 3-Year CAGR
- Sharpe Ratio
- Alpha
- Expense Ratio
- Maximum Drawdown



Each metric was ranked and assigned a weighted score. Funds with strong returns, favorable risk-adjusted performance and lower downside risk received higher rankings.

Benchmark analysis was performed using NIFTY 50 and NIFTY 100 indices. The comparison enabled evaluation of fund performance relative to broader market movements and helped identify schemes consistently outperforming benchmark returns.

Advanced Analytics

Advanced analytics techniques were implemented to evaluate investor behaviour, portfolio concentration and downside risk.

The following analyses were performed:

- Historical Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Rolling 90-Day Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Herfindahl-Hirschman Index (HHI) Concentration Analysis

Historical VaR and CVaR were calculated using daily return distributions to estimate potential downside losses. Rolling

Sharpe Ratios were used to evaluate changes in risk-adjusted performance over time.

Investor Cohort Analysis grouped investors based on their first transaction year to study long-term participation trends. SIP Continuity Analysis identified investors with irregular investment patterns, while HHI was used to measure portfolio concentration across mutual fund schemes.

These techniques provided deeper insights beyond traditional performance metrics and helped evaluate both investor behaviour and portfolio risk.

Investor Analytics

Investor transaction data was analyzed to understand investment behaviour and participation trends.

The analysis included:

- Transaction amount distribution
- State-wise investment activity
- Investor demographic analysis
- Transaction type comparison
- SIP participation patterns

The results indicated varying levels of investor participation across regions and investment categories. Transaction analysis highlighted the importance of SIP investments in driving mutual fund growth.

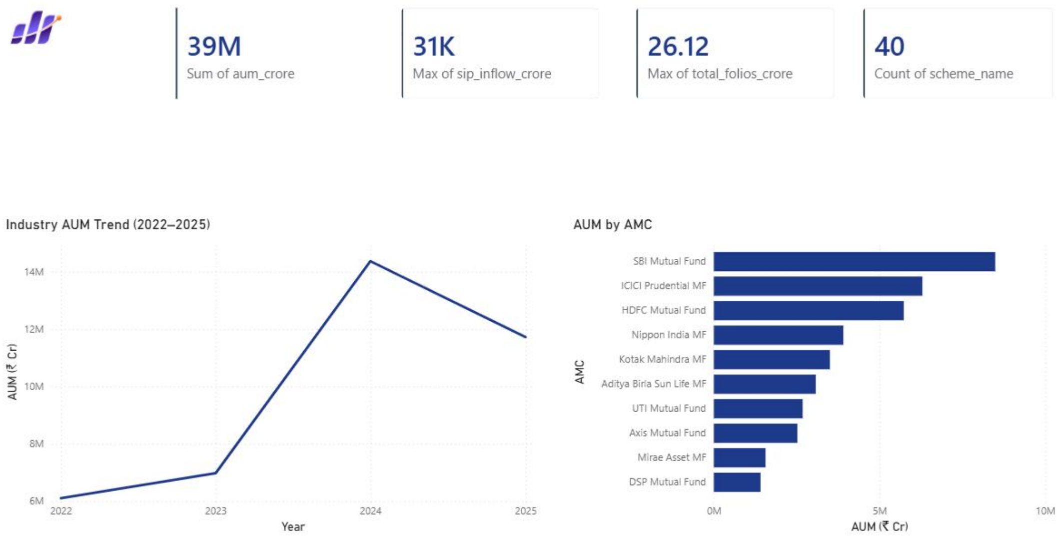
Investor analytics provided valuable insights into customer behaviour and supported the identification of long-term investment trends.

Dashboard Overview

An interactive Power BI dashboard was developed to present analytical findings in a visually intuitive manner.

The dashboard consists of multiple pages covering:

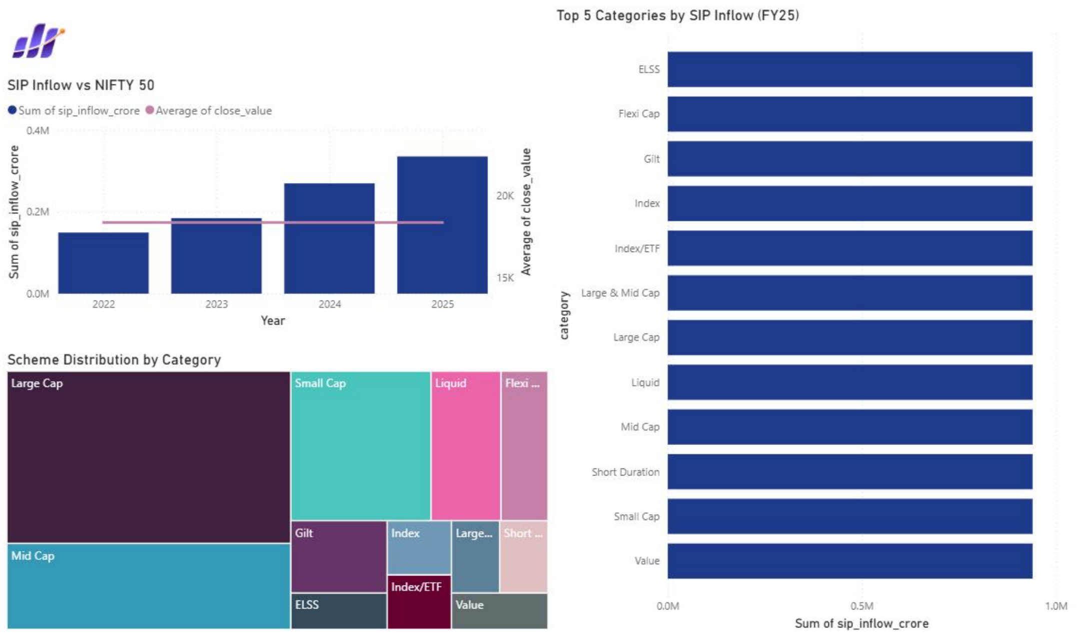
- Industry Overview



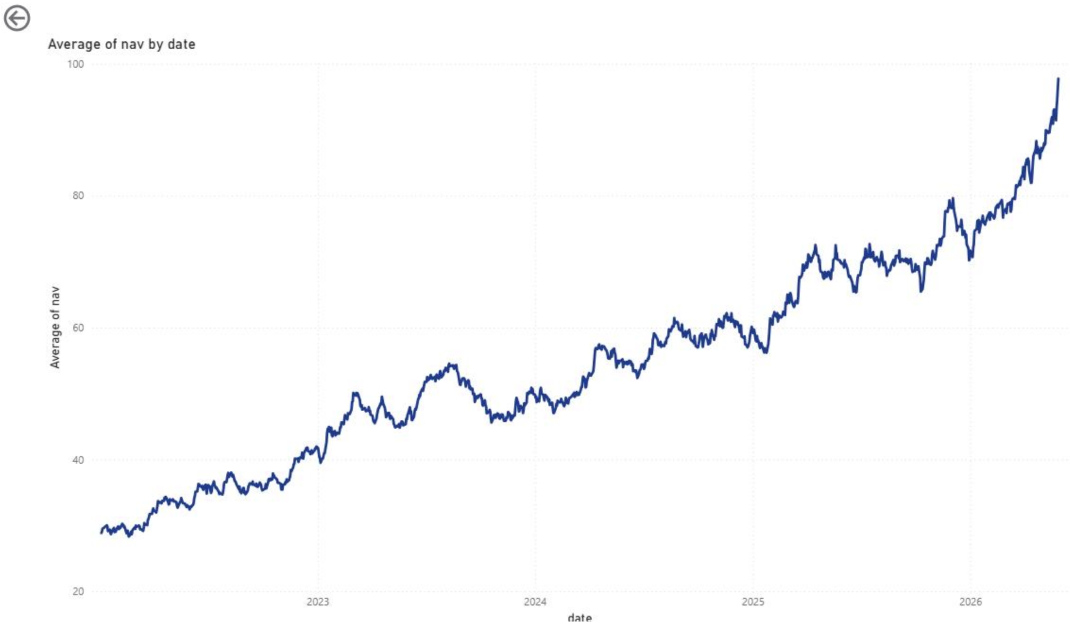
- Fund Performance Analysis

- # Investor Analytics

- # SIP and Market Trends



- NAV Trend Analysis



Interactive filters, KPI cards and visualizations were incorporated to enable dynamic exploration of mutual fund data. The dashboard allows users to compare schemes, analyze investor participation and monitor industry trends efficiently.

The dashboard serves as the final visualization layer of the analytics platform and transforms complex analytical outputs into actionable business insights.

Dashboard Insights

The dashboard revealed several important industry and investor trends.

Key observations include:

- Assets Under Management (AUM) showed sustained growth across the study period.
- SIP inflows demonstrated increasing retail investor confidence.
- Certain fund houses consistently maintained higher market share.
- Risk-adjusted performance varied significantly among schemes.
- Investor participation increased across multiple fund categories.

The dashboard provided a consolidated view of performance, risk and investor behaviour, enabling efficient monitoring of mutual fund industry dynamics.

Key Findings

The project generated several valuable insights from mutual fund and investor data.

Major findings include:

- Industry AUM and SIP inflows exhibited positive long-term growth trends.
- Risk-adjusted metrics such as Sharpe Ratio and Sortino Ratio provided better fund evaluation than absolute returns.
- Benchmark comparison identified schemes that consistently outperformed market indices.
- VaR and CVaR highlighted differences in downside risk exposure across funds.
- Investor cohort analysis revealed varying investment behaviour among different groups.
- Portfolio concentration levels differed significantly across schemes.

These findings demonstrate the importance of combining performance, risk and behavioural analytics when evaluating mutual funds.

Recommendations and Future Scope

Based on the analysis conducted, the following recommendations are proposed:

- Use risk-adjusted metrics alongside returns when comparing mutual funds.
- Monitor portfolio concentration to reduce exposure to sector-specific risks.
- Encourage SIP continuity to improve long-term investor participation.
- Regularly compare scheme performance against benchmark indices.

Future enhancements may include:

- Real-time NAV integration through APIs
- Machine learning-based return prediction
- Portfolio optimization models
- Advanced recommendation systems

Automated dashboard refresh mechanisms

These improvements can further enhance the analytical capabilities of the platform and support more informed investment decisions.

Conclusion

The Mutual Fund Analytics Platform successfully integrated data engineering, database management, financial analytics and business intelligence into a unified solution.

The project involved data collection, cleaning, database design, exploratory analysis, performance evaluation, advanced risk analytics and dashboard development. Multiple financial metrics were implemented to assess fund performance and risk exposure, while investor analytics provided insights into participation trends and investment behaviour.

The Power BI dashboard transformed analytical outputs into interactive visualizations, enabling efficient decision-making and trend analysis.

Overall, the project demonstrates the practical application of data analytics techniques in the mutual fund domain and provides a scalable foundation for future financial analytics solutions.